Wenjin Yang

(+86) 153-9077-7338 | csyangwenjin@outlook.com

PhD Application: AI4S / DS / AI / ML

Education

ShanghaiTech University, M.S. in Computer Science and Technology, GPA: 3.56/4.00 *Sep. 2022 – Jun. 2025*

- **Selected Coursework**: Algorithm Design and Analysis, Matrix Computations, Machine Learning, Deep Learning

ShanghaiTech University, B.S. in Computer Science and Technology, GPA: 3.01/4.00 *Sep. 2018 – Jun. 2022*

- **Selected Coursework**: Data Structures and Algorithms, Computer Vision, Operating Systems, Computer Organization

Research Interests

- Focus on two core challenges in AI4S: data scarcity and physical constraints. Research on structured representation learning from sparse time-series data and mechanisms for embedding physical priors, aiming to build battery foundation models that achieve cross-condition generalization and physical interpretability for system state prediction under extremely limited samples. Building on this, further explore learning and reasoning of latent state evolution toward inferable and extrapolatable modeling of battery degradation dynamics.

Research Experience

Early Battery Cycle Life Prediction via Assembled Capacity Curve, *Independent Lead Dec. 2023 – Jan. 2025*

- Addressed the challenge of sparse early-cycle data limiting the accuracy of traditional feature-engineering methods for battery life prediction. Proposed the **Assembled Capacity Curve** ($Q_a = Q_c - Q_d$) and designed a transformation pipeline converting the 1D curve into a three-channel 224×224 image, reformulating time-series prediction as an image regression task to extract degradation patterns from sparse data.
- Fine-tuned a pretrained CNN via transfer learning across three public datasets (MIT/TJU/XJTU). Achieved average MAPEs of **7.86%–15.27%**, reducing error by approximately **2**× over feature-engineering methods.
- Published in *Journal of Power Sources* (**JCR Q1, IF=8.1**).

Visual Representation of Time-Series Data for Deep Predictive Modeling, *Independent Lead Dec. 2023 – Jan. 2025*

- Identified the limitation of CNNs that only capture local patterns within kernel windows, losing global trend information. Designed a **time-series visualization** method that maps 1D signals into 2D structured images, enabling convolutions to capture both local details and global context while preserving physical semantics for interpretability.
- Achieved an average prediction error **below 7.1%** on the MIT public dataset, demonstrating ultra-early performance prediction capability from single-cycle visualized data for rapid battery screening. Published in *IEEE Transactions on Power Electronics* (**CAS Q1 Top, IF=6.6**) and led to a granted invention patent.

Battery SOH Prediction for On-Road EVs Using Field Data, *Main Contributor Oct. 2023 – Jul. 2024*

- Addressed the challenge of reliable battery capacity labeling from field data with low-resolution SOC measurements and transient current fluctuations. Proposed a charging process screening method that suppresses noise through SOC range constraints and steady-current fragment selection. Systematically extracted **21 user behavior features** characterizing driving and charging patterns, and employed **LSTM** to model the temporal dependencies between behavioral features and capacity degradation.
- Validated on 200 on-road EVs from the Shanghai EVData center; achieved an average MAE of **0.91%** and RMSE of **1.12%** on the test set. Published in *Proc. ITEC 2024* and led to a granted invention patent.

Publications

Google Scholar: 52 citations, h-index 5 (August 2026)

Journal Papers

- **Wenjin Yang**, Fanqi Min, Jingying Xie, and Hengzhao Yang, "Ultra-early prediction of lithium-ion battery cycle life based on visualized single-cycle data," *IEEE Transactions on Power Electronics*, vol. 40, no. 5, pp. 7342–7353, 2025. (**CAS Q1 Top, IF=6.6**)
- **Wenjin Yang** and Hengzhao Yang, "Ultra-early prediction of lithium-ion battery cycle life based on assembled capacity curve extracted from a single cycle," *Journal of Power Sources*, vol. 640, pp. 236620:1–236620:12, 2025. (**JCR Q1, IF=8.1**)
- Pengyu Wang, **Wenjin Yang**, and Hengzhao Yang, "An ultra-early prediction framework for battery capacity degradation trajectories using visualized single-cycle data," *Batteries & Supercaps*, vol. 9, no. 8, p. e70427, 2026.

Conference Papers

- **Wenjin Yang** and Hengzhao Yang, "Charging pricing incentives-enabled coordinated dispatching for improved overall benefits of electric vehicles and islanded photovoltaic charging stations," in *Proc. EPE*, 2023, pp. 1–7.
- Yifei Xu, Yingjie Zhang, **Wenjin Yang**, Caiying Xu, Fuhao Wang, and Hengzhao Yang, "A battery state of health prediction framework considering user behavior for on-road electric vehicles," in *Proc. ITEC*, 2024, pp. 1–6.
- Can Zhang, Caiying Xu, **Wenjin Yang**, Yifei Xu, Mingxia Wu, Zhongxun An, and Hengzhao Yang, "An online supercapacitor capacitance labeling method based on tram field data," in *Proc. ITEC*, 2024, pp. 1–5.
- Jiaqi Zhao, Song Feng, Caiying Xu, Yifei Xu, **Wenjin Yang**, and Hengzhao Yang, "A multi-fault diagnosis framework for electric vehicle batteries," in *Proc. ITEC+EATS*, 2025, pp. 1–7.
- Yuhong Fan, Song Feng, Yifei Xu, Jiaqi Zhao, Caiying Xu, **Wenjin Yang**, and Hengzhao Yang, "A self-discharge fault diagnosis method for electric vehicle batteries," in *Proc. PEDG*, 2025, pp. 605–609.

Patents

- Hengzhao Yang, **Wenjin Yang**, "State evaluation method and related apparatus for heterogeneous uninterruptible power supply clusters," Chinese Invention Patent, ZL202510844074.1, 2025.
- Hengzhao Yang, **Wenjin Yang**, "Battery life prediction method based on visualized single-cycle sampling data and related apparatus," Chinese Invention Patent, ZL202411882164.1, 2024.
- Hengzhao Yang, Jiaqi Zhao, Caiying Xu, Yifei Xu, **Wenjin Yang**, "A hierarchical identification-based fault diagnosis method for vehicle power batteries," Chinese Invention Patent, ZL202411634580.X, 2024.
- Hengzhao Yang, Can Zhang, Caiying Xu, Tingli Huang, Zhongxun An, Mingxia Wu, **Wenjin Yang**, Yifei Xu, "Capacitance calibration method for supercapacitor power systems in vehicles," Chinese Invention Patent, ZL202410615364.4, 2024.
- Hengzhao Yang, Yingjie Zhang, Yifei Xu, Fuhao Wang, **Wenjin Yang**, Caiying Xu, "User-behavior-based battery state of health prediction method and system," Chinese Invention Patent, ZL202410615365.9, 2024.

Additionally, 2 pending invention patents and 3 software copyrights.

Professional Experience

CATL, 21C Innovation Lab, Algorithm Research *Jul. 2025 – Present*

- Develop time-series forecasting and anomaly detection algorithms for production-scale industrial data, covering the full pipeline from data cleaning and feature engineering to model selection and offline validation.
- Independently led an AI algorithm project, covering the full process from problem definition and solution design to algorithm validation.
- Contributed to interdisciplinary research bridging deep learning and scientific data analysis for practical industrial deployment.

Honors and Awards

- **National Second Prize**, "Challenge Cup" (National Level)
- **Third Prize**, Energy Electronics Industry Innovation Competition (National Level)

- Digital Car Competition (Battery Life Prediction) — **Top 100/1252**
- **Outstanding Graduate of ShanghaiTech University**, 2025
- ShanghaiTech University Second-Class Scholarship, 2022–2025 (three consecutive years)

Research Skills

- **Machine Learning**: time-series analysis, deep representation learning, physics-informed deep learning, Bayesian optimization
- **Programming & Tools**: Python, PyTorch, C/C++, Git, LaTeX
- **Languages**: Chinese (native), English (CET-6: 542, academic writing and presentation)